

ITA0608 - Machine Learning

List of Lab Exercises

10. Write a Python Program to Implement Expectation & Maximization Algorithm

Aim

To implement the Expectation-Maximization (EM) Algorithm for clustering data using a Gaussian Mixture Model (GMM) in Python.

Algorithm

1. Import the required libraries.
2. Load a dataset (Iris dataset).
3. Select numerical features for clustering.
4. Standardize the dataset.
5. Initialize the Gaussian Mixture Model with the desired number of clusters.
6. Train the model using the Expectation-Maximization (EM) algorithm.
7. Predict the cluster labels.
8. Display the cluster assignments.
9. Visualize the clustered data.

Program Code

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from sklearn.preprocessing import StandardScaler

from sklearn.mixture import GaussianMixture

# Load Iris Dataset

iris = load_iris()

# Create DataFrame
```

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)

# Standardize the data

scaler = StandardScaler()

X = scaler.fit_transform(df)

# Apply Gaussian Mixture Model (Expectation Maximization)

gmm = GaussianMixture(

    n_components=3,

    random_state=42

)

gmm.fit(X)

# Predict Cluster Labels

clusters = gmm.predict(X)

# Add Cluster Labels to DataFrame

df["Cluster"] = clusters

# Display first 10 records

print("First 10 Records with Cluster Labels:\n")

print(df.head(10))

# Plot Clusters

plt.figure(figsize=(8,6))

plt.scatter(

    X[:,0],

    X[:,1],

    c=clusters,

    cmap='viridis',
```

```
s=60

)

plt.title("Expectation Maximization Clustering")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.colorbar(label="Cluster")
plt.show()
```

Output

```

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.mixture import GaussianMixture

# Load Iris Dataset
iris = load_iris()

# Create DataFrame
df = pd.DataFrame(iris.data, columns=iris.feature_names)

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```

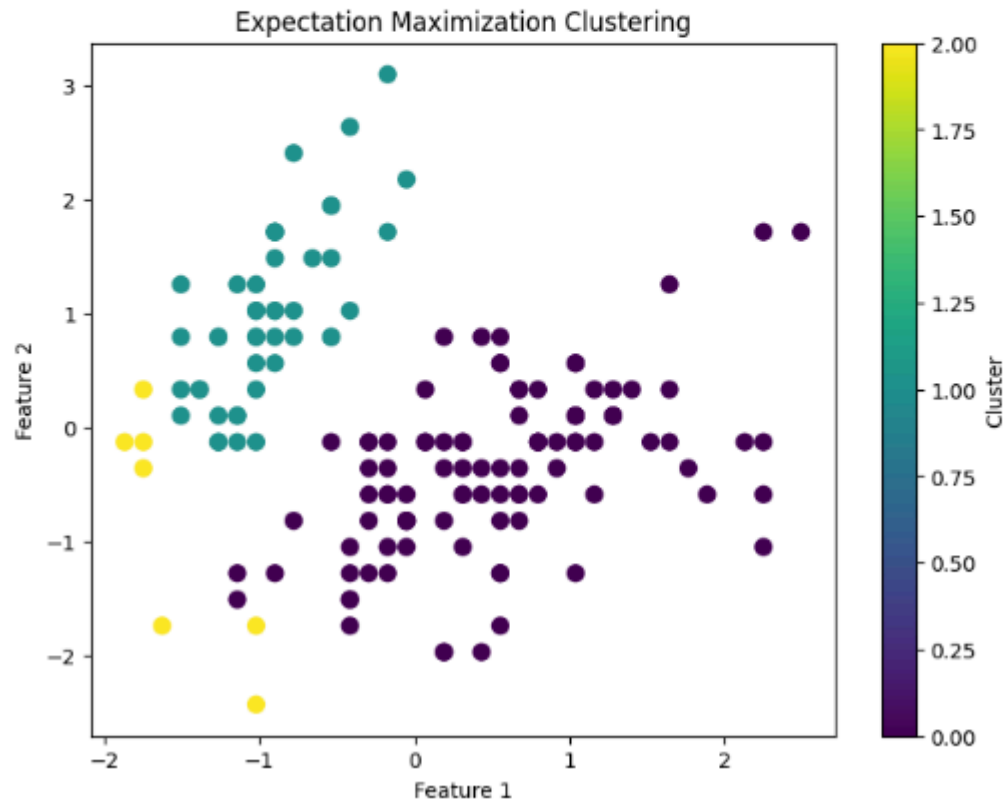
```

...
    sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                    5.1             3.5             1.4             0.2
1                    4.9             3.0             1.4             0.2
2                    4.7             3.2             1.3             0.2
3                    4.6             3.1             1.5             0.2
4                    5.0             3.6             1.4             0.2
5                    5.4             3.9             1.7             0.4
6                    4.6             3.4             1.4             0.3
7                    5.0             3.4             1.5             0.2
8                    4.4             2.9             1.4             0.2
9                    4.9             3.1             1.5             0.1

    Cluster
0         1
1         1

```

	cluster
0	1
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	2
9	1



Result

The Expectation-Maximization (EM) Algorithm was successfully implemented using the Gaussian Mixture Model (GMM). The model clustered the Iris dataset into three groups, and the cluster assignments were displayed and visualized successfully.